DIFF: task requirements and completed code determine the next subgoal

Boxed token = query site at the pivot. Shaded instruction = task K,V span. Dashed arrow = activation transplant.

(a) Read what remains

A goal before the next condition is written

CLEAN PROMPT

Keep words that are palindromes
and start with a vowel.

```
def func(words):
    res = [w for w in words
           if w == w[::-1]]
    return [w for w in res if
```

At the boxed if token:

Palindrome filtering is already in the code.

The remaining condition is vowel-initial.

Goal-head attention reads the task span.

CLEAN CONTINUATION

```
w[0] in "aeiou"]
```

GOALSCOPE · SUMMARY

Check whether the word starts with a vowel.

The DIFF account

Q at the pivot: completed work

K,V in the instruction: task requirements

Head output: the selected next subgoal

(b) Transplant progress

Same task; different completed code

SOURCE · vowel condition completed

Keep words that are palindromes
and start with a vowel.

```
def func(words):
    res = [w for w in words
           if w[0] in "aeiou"]
    return [w for w in res if
```

Cache Q(if) → overwrite destination Q(if)

DESTINATION · palindrome condition completed

Keep words that are palindromes
and start with a vowel.

```
def func(words):
    res = [w for w in words
           if w == w[::-1]]
    return [w for w in res if
```

CLEAN NEXT CONDITION

```
w[0] in "aeiou"]
```

PATCHED NEXT CONDITION

```
w == w[::-1]]
```

Readout: vowel check → palindrome check

The destination code prefix is unchanged.

(c) Transplant requirements

Different tasks; same completed code

SOURCE · shortest word / vowel

Return whether the shortest word
starts with a vowel.

```
def func(words):
    res = sorted(words, key=len)
    return
```

Cache K,V over the shaded source instruction; overwrite K,V at the destination instruction.

DESTINATION · longest word / palindrome

Return whether the longest word
is a palindrome.

```
def func(words):
    res = sorted(words, key=len)
    return
```

CLEAN CONTINUATION

```
res[-1] == res[-1][::-1]
```

PATCHED CONTINUATION

```
res[0][0] in "aeiou"
```

Readout: palindrome check → vowel check

The sorting prefix and pivot stay in place.

Adapted from goal-diff slides 1, 3 and 5. Prompts, code and readouts are shortened schematic examples, not verbatim model outputs.
Nonempty lowercase words assumed. The arrow in (a) is schematic; patch duration, head sets and span alignment belong in the caption.